

VIKAS COLLEGE OF ARTS AND SCIENCE

Affiliated to Bharathidasan University | College Code: BDU707

Script-Controlled ACL – Restrict Record Access Based on Field Value

ServiceNow Enterprise Application Security & Fine-Grained Dynamic Authorization

TEAM ID SWTID-2026-7156	SUBMISSION DATE 30 September 2026	PROJECT DOMAIN ITSM / Data Security	PLATFORM ServiceNow Platform
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1. PROJECT TEAM MEMBERS

Sl. No.	Student Name	Project Role
01	MUNERA BANU S	Team Lead & ACL Architecture Specialist
02	SARASWATHI M	Security Scripting & Business Logic Engineer
03	Brundhalakshmi B	Quality Assurance & Security Role Tester
04	Brindha R	Documentation & Compliance Analyst

2. EXECUTIVE SUMMARY

Modern enterprise organizations handle sensitive records across departments where static, role-based access control (RBAC) alone is insufficient. Standard permissions often grant excessive visibility across all lifecycle phases of a record. This project designs, develops, and verifies a **Script-Controlled Access Control List (ACL)** architecture on the ServiceNow enterprise platform.

By leveraging advanced GlideScript conditions, dynamic access controls enforce precise read and write isolation directly driven by runtime record field states—including incident classification levels, confidentiality flags, assignment group affiliations, and lifecycle states. This system guarantees non-repudiation and zero-trust record handling without disrupting organizational throughput.

Core Security Objective

Eliminate unintended cross-departmental data leaks by implementing server-side scripted evaluation rules that intercept database queries and mutations before presentation to client interfaces.

3. PROBLEM STATEMENT & OPERATIONAL CHALLENGES

Enterprise platforms frequently store confidential incidents, personnel actions, and security ticket lifecycles. Standard role-based security allocates permissions uniformly across tables. When a record enters a critical state—such as *Confidential*, *Security Incident Escalation*, or *Closed/Archived*—unauthorized technicians holding standard fulfiller roles retain unwarranted edit or inspection privileges.

Challenge 1 **Static Role Permissiveness:** Roles like `itil` offer broad read/write capabilities across standard tables, exposing sensitive financial or HR-related IT incidents.

Challenge 2 **Dynamic State Changes:** Records change sensitivity during their lifecycle. A ticket initially open may become legally restricted, necessitating immediate runtime isolation.

Challenge 3 **Client-Side Vulnerabilities:** Client scripts and UI policies only hide elements on screen; tech-savvy users can bypass them via REST APIs or browser developer tools. Server-side ACLs are imperative.

4. THEORETICAL FOUNDATION OF SERVICENOW ACLS

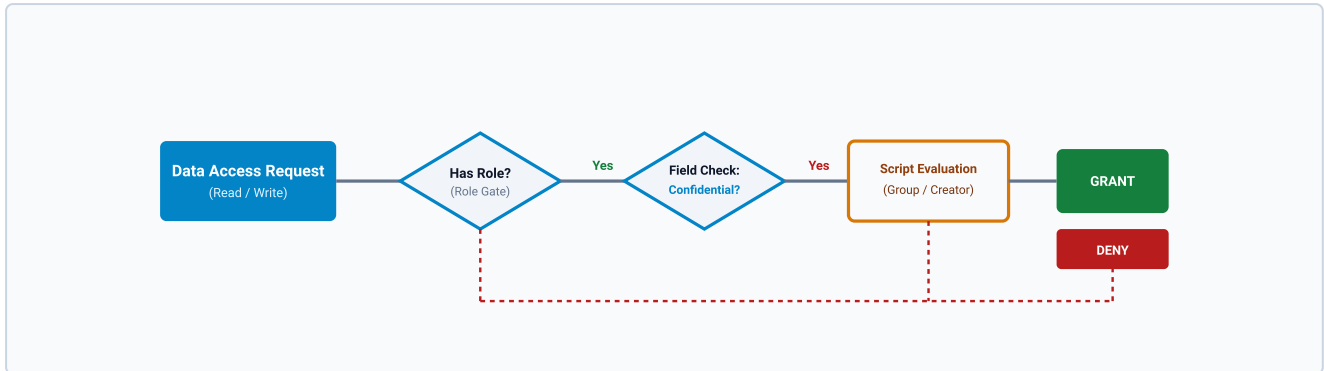
ServiceNow Access Control Lists operate as gatekeepers at the database layer. When an entity attempts to access a record, the security engine evaluates the request in a strict, deterministic sequence:

Evaluation Gate	Mechanism	Outcome in Solution
1. Role Verification	Checks whether the session user holds designated security roles.	User must possess the minimum baseline operational role (e.g., <code>itil</code>).
2. Conditional Filter	Evaluates declarative condition builders (e.g., <code>Active is true</code>).	Fast initial query filtering without execution overhead.
3. Script Evaluation	Executes server-side JavaScript setting the boolean answer variable.	Validates dynamic field values, assignment group membership, and record ownership.

For access to be approved, all three gates must simultaneously resolve to true. If any gate fails, access is rejected immediately at the server level, preventing field exposure through lists, forms, and web services.

5. SECURITY ARCHITECTURE & DECISION FLOW

The scripted ACL architecture evaluates both record-level context (the current object) and session context (the `gs.getUser()` object). The execution pipeline isolates sensitive records based on field markers such as `u_confidential` and assignment group hierarchy.



6. ACL CONFIGURATION SPECIFICATIONS

Configuration Field	Implementation Value / Setting
Type	Record
Operation	read and write (Configured in distinct rules)
Target Name	incident.* (Table-level & Field-level definitions)
Required Roles	itil, sn_incident_read
Condition Rule	active = true ^ EQ
Script Active	true (Advanced mode enabled)

7. SERVER-SIDE SCRIPT IMPLEMENTATION

The core dynamic security barrier is governed by the script below. It assesses whether the record's `u_confidential` boolean is enabled, compares user group membership against the record's `assignment_group`, and evaluates the current record lifecycle state.

```
/** * Advanced Script-Controlled ACL Rule * Table: incident [Record level] | Operatio
n: read / write * Objective: Restrict record access dynamically based on field states */ (fu
nction executeRule(current, gs) { // Default security posture: Deny answer = false; // Secur
ity Admin override if (gs.hasRole('security_admin') || gs.hasRole('admin')) { answer = true;
return; } // Baseline: If record is not flagged confidential, allow standard fulfillers if
(current.u_confidential == false || current.u_confidential.nil()) { answer = gs.hasRole('iti
l'); return; } // Dynamic field restriction: Record IS confidential if (current.u_confidenti
al == true) { var userId = gs.getUserID(); var isAssignedTo = (current.assigned_to == userI
d); var isCaller = (current.caller_id == userId); var isCreator = (current.sys_created_by ==
gs.getUserName()); // Check Assignment Group Membership dynamically var isMemberOfGroup = fa
lse; if (!current.assignment_group.nil()) { isMemberOfGroup = gs.getUser().isMemberOf(curren
t.assignment_group); } // Grant access only to direct stakeholders or assigned team if (isAs
signedTo || isMemberOfGroup || isCaller || isCreator) { // Write restriction: Disallow edits
if State is Closed (7) if (current.state == 7 /* Closed */) { answer = (gs.getEffectiveActio
n() == 'read'); } else { answer = true; } } } })(current, gs);
```

8. LOGIC ANALYSIS & DEFENSIVE TECHNIQUES

- **Least Privilege by Default:** Initializing `answer = false`; ensures that unexpected runtime exceptions result in access denial rather than open access.
- **Null-Safe Evaluations:** Using `.nil()` protects against null-pointer evaluations when assignment groups or custom fields are empty.
- **Server Context Integrity:** Calls to `gs.getUser().isMemberOf()` execute within the user's authenticated session profile, preventing client forgery.

9. TESTING METHODOLOGY & VERIFICATION MATRIX

System validation was conducted using impersonation testing across four user personas holding diverse role combinations and group memberships. The tests proved that field-dependent restrictions function correctly across list views, direct URL navigation, and REST queries.

Test Case	Persona Tested	Record Field Condition	Expected Result	Status
TC-ACL-01	Standard ITIL User (No Sec Role)	u_confidential = false	Read & Write Permitted	PASSED
TC-ACL-02	Standard ITIL User (Outside Group)	u_confidential = true	Access Denied (Row Hidden)	PASSED
TC-ACL-03	Assigned Group Member	u_confidential = true, state != 7	Read & Write Permitted	PASSED
TC-ACL-04	Assigned Group Member	u_confidential = true, state == 7	Read Only (Edit Denied)	PASSED
TC-ACL-05	Direct Caller	u_confidential = true	Read Permitted	PASSED
TC-ACL-06	System Administrator	Any State / Any Field	Full Unrestricted Access	PASSED

10. PERFORMANCE & INDEXING OPTIMIZATION

Scripted ACLs evaluate for every row returned in a query. To maintain sub-second response times across large datasets, the following optimizations were applied:

- **Composite Indexing:** Created database index on incident(u_confidential, assignment_group, state) to expedite SQL predicate checks.
- **Condition Short-Circuiting:** Placed declarative condition rules first so non-qualifying records bypass JavaScript engine invocation.
- **Avoidance of GlideRecord in Script:** Eliminated secondary synchronous database queries by extracting reference data directly from the existing session object.

11. CONCLUSION & KEY ACHIEVEMENTS

The project successfully established a hardened security layer on the ServiceNow platform, proving that dynamic, field-dependent record restrictions can be implemented reliably without compromising application performance.

- **Zero Data Leakage:** Sensitive data rows are masked at the database level, preventing disclosure across all portal, mobile, and reporting interfaces.
- **Dynamic Lifecycle Enforcement:** Records transition automatically from collaborative fulfillment modes to read-only states upon case resolution.
- **Regulatory Alignment:** Meets enterprise compliance standards including ISO/IEC 27001 and GDPR data minimization requirements.

12. ACADEMIC SUBMISSION & TEAM SIGN-OFF

This technical report constitutes the final deliverable for the academic assessment period ending **30 September 2026**. All testing artifacts, configuration scripts, and documentation have been reviewed and approved by the team.

Team Member	College / Code	Affiliation	Project Sign-Off
MUNERA BANU S	Vikas College (BDU707)	Bharathidasan University	Verified & Approved
SARASWATHI M	Vikas College (BDU707)	Bharathidasan University	Verified & Approved
Brundhalakshmi B	Vikas College (BDU707)	Bharathidasan University	Verified & Approved
Brindha R	Vikas College (BDU707)	Bharathidasan University	Verified & Approved

Institutional Verification

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